The same force can produce different accelerations

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Inverse relationship / ES: relación inversa / PT: relação inversa

Double M halves a at fixed net force.

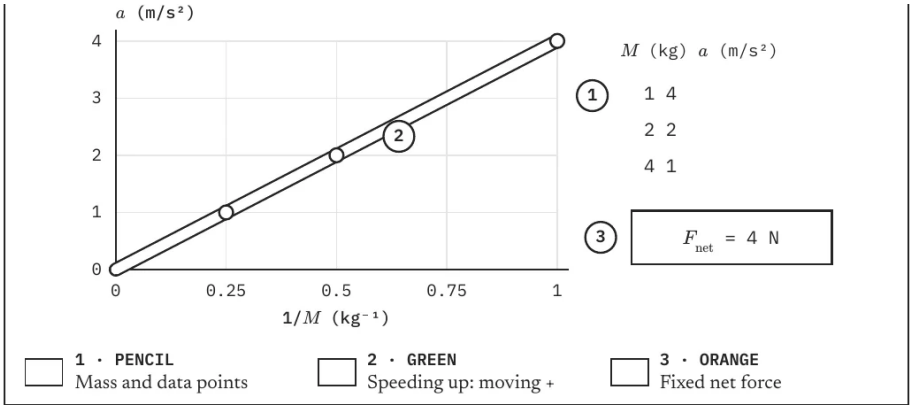
Also on your note page: System mass, Control, Slope.

B READ THE RELATIONSHIP

a = Fnet/M; the slope of a versus 1/M is Fnet

As you read, find: change · control · compare.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Under the same net force, a 1 kg system accelerates at 4 m/s² and a 2 kg system at 2 m/s² . Predict a for a 3 kg system.

Given M 1 = 1 kg ; a 1 = 4 m/s² M 2 = 2 kg ; a 2 = 2 m/s² same net force

Rule Check M × a . Then a = F net / M .

Work 1(4) = 2(2) = 4 N a 3 = 4 N ÷ 3 kg

Answer 1.33 m/s² Assumes net force stays 4 N .

E SAVE FOR THE FINAL MEETING

For the same force, a more massive XJ changes velocity less quickly. The comparison needs the same net force; a larger gravitational force can change that result.

Read the physics: Inverse relationship

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Compare a light cart with a loaded cart. Use the same net force and the same release method. Otherwise a change in acceleration could have more than one cause.

02 HOW IT WORKS

Read acceleration from the slope of a velocity–time graph. At fixed net force, doubling the moving mass halves acceleration. Multiplying mass by acceleration gives the same net force in both trials.

03 WHAT TO CHECK

Keep the hanging mass, track and release conditions consistent. Name what changed and what was controlled before stating the result.

04 TRY THE EXAMPLE FROM YOUR NOTES

Under the same net force, a 1 kg system accelerates at 4 m/s² and a 2 kg system at 2 m/s² . Predict a for a 3 kg system. 1.33 m/s² Assumes net force stays 4 N .

Model check: a = Fnet/M; the slope of a versus 1/M is Fnet.

05 CONNECT TO XJ

For the same force, a more massive XJ changes velocity less quickly. The comparison needs the same net force; a larger gravitational force can change that result.

SAY IT BEFORE YOU WRITE

Which quantity must match in the two trials? We changed ___. We kept ___ fixed. Acceleration ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Inverse relationship / ES: relación inversa / PT: relação inversa

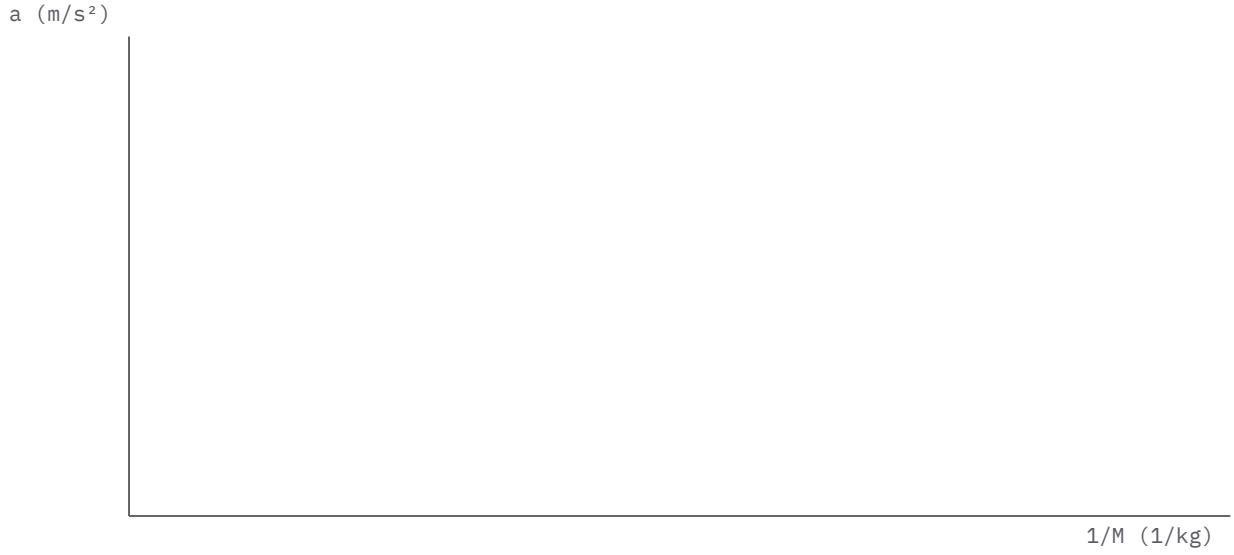
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, inverse relationship means …	Keep the net force fixed. If the mass doubles, what happens to acceleration? When mass doubles, acceleration ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Sketch a versus 1/M. Label the axes and units, draw the rising line, and name the quantity represented by its slope.

Labels you can use: change · control · compare.



Tell a partner: We changed ___. We kept ___ fixed. Acceleration ___.

On assignment: Inverse relationship

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two asteroid models get the same net force for the same time. Model H has twice the mass of L.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Test engineer**

Build a two-row test card. Pick the model with the larger velocity change and show the mass comparison for the test crew.

☐ **Lab correspondent**

Explain to viewers why two equal pushes can look different. Your producer must check what stayed the same in both trials.

Word bank: change · control · compare.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”

Editor’s check: Use mass × acceleration as an independent check.